

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2020
URP 3291
(Operations Research and System Analysis)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for any missing data.
iii) Answer any two questions from each section in separate script.

Section – A

- 1 (a) what is meant by Operation Research (OR)? Briefly discuss how OR can help a manager to take effective decision. 15
- (b) Solve the following problem using "linear programming" method. 15
Minimize, $Z = 5x_1 + 3x_2$
Subject to,
 $2x_1 + 4x_2 \geq 16$
 $4x_1 + 3x_2 \geq 24$
 $x_1, x_2 \geq 0$
- 2 Solve the following problem by Simplex method 30
Maximize, $Z = 10x_1 + 4x_2$
Subject to,
 $4x_1 + 5x_2 \leq 100$
 $5x_1 + 2x_2 \leq 80$
 $x_1, x_2 \geq 0$
- 3 Find the Saddle point and the value of the game 30

	I	II	III	IV
I	4	-4	-5	6
II	-3	-4	-9	-2
III	6	7	-8	-9
IV	7	3	-9	5

Section – B

- 4 People are infected COVID-19 virus at a rate of one every 6 minute in Bangladesh. The time between infections follows exponential distribution. Find 30
(i) The average number of infections per month.
(ii) The probability that no infection will occur in any one day.
- 5 (a) Why the mean arrival rate must be less than the mean service rate of a queuing system? Explain. 10
(b) Consider the following payoff (cost) matrix: 20

	b_1	b_2	b_3	b_4	b_5
a_1	15	0	-6	10	12
a_2	9	14	3	8	-2
a_3	5	1	20	-10	7
a_4	18	10	2	12	5

No probabilities are known for the occurrence of future states. Determine the solution by the following criteria.

- (i) Laplace, (ii) Minimax, (iii) Savage, (iv) Herwicz, (Assume, $\alpha = 0.06$)

- 6 (a) Consider a Markov chain with two states and transition probability matrix,

12

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$

In the long run, what proportion of the time is the process in each of the two states?

- (b) Two white and two black balls are distributed in two urns in such a way that each contains two balls. We say that, the system is in state i , $i = 0, 1, 2$, if the first urn contains i black balls. At each step, we draw one ball from each urn and place the ball drawn from the first urn into the second, and conversely with the ball from the second urn, write down the detailed calculation to find its transition probability matrix.

18

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP3rdYear 2ndTermRegularExamination, 2020
URP 3231
(Transportation Planning)

Full Marks: 120

Time: 1:30 hrs

- N.B.** i) Answer any three questions from each section in a separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) You traveled to your home from KUET Campus several times. Discuss in brief (12)
the components of the transportation system that you frequently use for the
above purpose. Do you think the system is perfect? If so, describe the strength
of the system. If not, explain why?
- (b) Assume, you are a KCC transportation planner and you have noticed that the (18)
number of e-bike accidents in seven wards in relation to the number of road
intersections is as follows:

No. of e-bike accidents	27	57	63	5	110	60	18
No. of road intersections	10	11	15	9	2	16	12

Set up a linear regression equation for e-bike accidents in terms of road intersections and also determine R^2 .

2. (a) The COVID-19 pandemic has had a significant impact on people's travel mode (10)
choice behavior. Is this true in your neighborhood? If so, discuss the influential
factors. If not, explain why?
- (b) Five Thana of Khulna City has the following information. (20)

TAZ	Thana Name	Area (sq. km)	Population	Production	Attraction
1	Kotwali	9.45	250651	35807	45350
2	Khan Jahan Ali	1.07	5177	740	600
3	Khalishpur	11.47	235018	33574	21681
4	Daulatpur	10.38	107573	15368	16559
5	Sonadanga	8.42	172079	24583	25881

The travel matrix is:

TAZ	Travel Cost (BDT)				
	1	2	3	4	5
1	10	50	35	25	15
2	50	15	30	25	40
3	35	30	5	10	30
4	25	25	10	5	25
5	15	40	30	25	10

Travel Cost (BDT)	Fij	Travel Cost (BDT)	Fij
5	7.45	30	2.21
10	5.50	35	1.05
15	4.33	40	0.75
25	2.51	50	0.62

Apply gravity model to distribute the trips ($K_{ij} = 3.85$).

3. (a) Suppose, you are responsible for planning a new railroad from Khulna to Mongla port. What basic elements of transportation planning will you consider? Discuss in brief with necessary examples. (15)
- (b) Why are travel demand forecasting models sequential and necessary for transportation planning? Discuss in brief the different demand forecasting models with examples of your community. (15)

Section – B

4. (a) Write a short note on Transport Demand Management (TDM) vs. Transportation System Management. (10)
- (b) Discuss the factors that are responsible to cause road accidents in brief. (10)
- (c) Exemplify any two of the following terms: (10)
- (i) Parking load (ii) Parking index (iii) Parking turnover
5. (a) How do we determine the pedestrian circulation in spaces? Consider a sidewalk next to your residence. If there are 1400 walkers per hour westward and 2000 eastward, operating at a level of service D/E (the break-point between LOS D and LOS E). What would be the linear pedestrian capacity of the sidewalk (width 5 feet)? (15)
- (b) What are the indicators of operational performance of urban bus service? In the case of Khulna City give your observation on each of the indicators. (15)
6. (a) Mention the principles of bus route planning. Illustrate different types of bus routes with necessary sketches. (15)
- (b) Make a comparative overview among curb-side bus stop, bus stop at a lay-by and bus cape with necessary sketches. (15)
-

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rdYear 2ndTerm Examination, 2020.
URP 3261
(Water Resources Planning)

Full Marks: 120

Time: 1.50hrs

- N.B.** i) Answer any TWO questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is watershed? Draw schematic sketch of watershed for any river? (06)
- (b) A catchment of area 140 km^2 received 125 cm rainfall in a year. At the outlet of the catchment, the flow in the stream draining the catchment was found to have an average rate of $2.0 \text{ m}^3/\text{s}$ for 3 months, $3.0 \text{ m}^3/\text{s}$ for 5 months and $6.0 \text{ m}^3/\text{s}$ for 4 months. (12)
 - (i) What is the runoff co-efficient for the catchment?
 - (ii) If afforestation of the catchment reduced the runoff co-efficient to 0.55, what is the increase in the abstraction from precipitation due to infiltration from precipitation due to infiltration, evaporation and transpiration, for the same annual rainfall of 125 cm ?
- (c) Analysis of data on maximum one-day rainfall depth at KUET indicated (12) that, a depth of 280 mm had a return period of 40 years. Determine the probability of one day rainfall depth equal to or greater than 280 mm at KUET occurring
 - (i) once in 20 successive years
 - (ii) two times in 15 successive year
 - and (iii) at least once in 20 successive years.

2. (a) Define unit hydrograph. What are the assumptions that constitute the foundations for the unit-hydrograph theory? (04)
- (b) Two storms of each 6-h duration and having rainfall excess values of 3.5 cm and 2.5 cm , respectively occurs successively. The 2.5 cm ER rain follows the 3.5 cm rain. The 6-h unit hydrograph for the catchment is given below. (12)

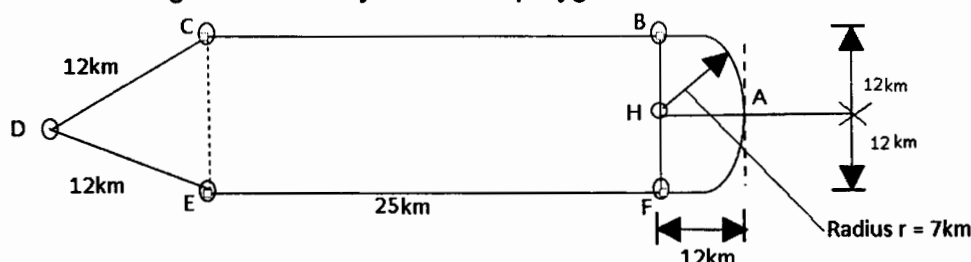
Time (h)	0	6	12	18	24	30	36	42	48
UH ordinate (m^3/s)	0	50	120	185	110	60	36	16	0

Calculate the resulting direct runoff hydrograph (DRH)?

- (c) Given the ordinates of a 4-h unit hydrograph as below, derive the ordinates of a 12-h unit hydrograph for the same catchment. (14)

Time (h)	0	4	8	12	16	20	24	28	32	36
Ordinates of 4-h UH (m^3/s)	0	20	85	140	170	130	90	38	12	0

3. (a) What is meant by hydrology and hydrological cycle? Describe the cycle with neat sketch. (18)
- (b) What is water budget of hydrological cycle? Find the mean precipitation for the area given below by Thiessen polygon: (12)



Section – B

4. (a) What is design flood and crop period? Suppose, 190 Floods have occurred in 150 years of records. It is desired to find out the chance of a flood having a frequency of 1000 years. (05)
- (b) What will happen when flood flow passes through the bend? Describe briefly. (12)
- (c) What is bathymetry survey? For a river the estimated flood of a two return periods are given below: (13)

Return Periods (years)	Peak Flow (m ³ /s)
150	550
75	477

What flood discharge in this river will have a return period of 1500 years? Use Gambler's method.

5. (a) What is irrigation? Why is it needed? What are the effects of irrigation? (10)
- (b) A stream of 150 lit/sec was diverted from a canal and 120 lit/sec were delivered to the field. An area of 2.0 hectars was irrigated in 7 hr. The effective depth of root zone is 1.8m. The runoff loss in the field was 450m³. The depth of water penetration varied linearly from 1.8m at the head end of the field to 1.2m at the tail end. Available moisture holding capacity of soil is 25cm per m depth of soil. Irrigation was started at a moisture extraction level of 45% of available moisture. Determine η_c , η_a , η_s and η_d . (10)
- (c) Write short notes on: (10)
- (i) Vegetation (ii) Guide banks (iii) Artificial cut off
6. (a) What are the quality parameters for irrigation water? Classify water according to SAR value and their use in irrigation. (12)
- (b) Explain the following water distribution technique with figure: (08)
- (i) Free flooding
- (ii) Drip irrigation method
- (c) What is the classification of irrigation water having the following characteristics: (10)
- Concentration of Na, Ca and Mg are 22, 3 and 1.5 mili-equivalents per liter respectively, and the electrical conductivity is 350 micro.ohms per cm at 25°C.